Supplementary Material

Enhancement of ammonia combustion with partial fuel cracking strategy: Laminar flame propagation and kinetic modeling investigation of $NH_3/H_2/N_2$/air mixtures up to 10 atm

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S1. Validation against laminar burning velocity data in literature

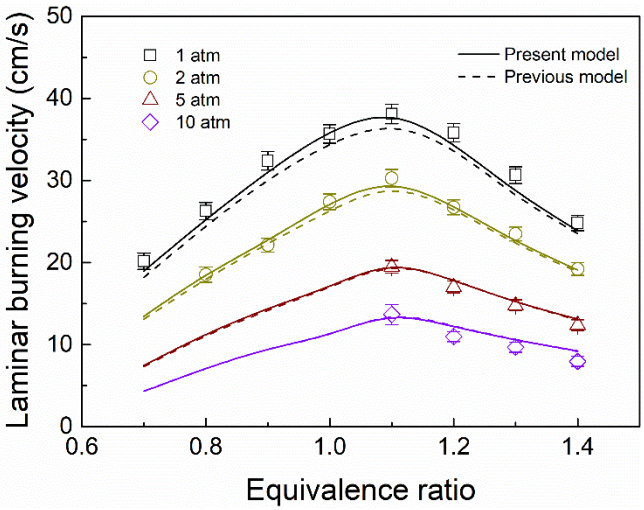


Figure S1 Comparison of the measured LBVs of partially cracked NH_3/air mixtures versus ϕ at γ = 40% and various P_u (symbols) with the simulated results using the present model (solid lines) and previous NH_3 model of our group [1] (dash lines).

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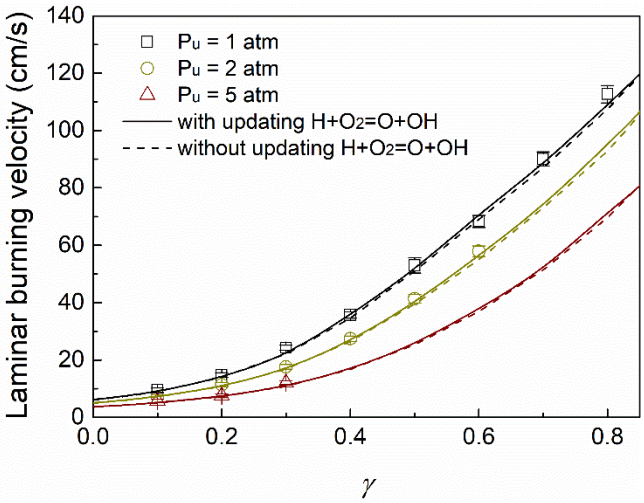


Figure S2 Comparison of the measured LBVs of partially cracked NH_3/air mixtures versus γ at ϕ = 1.0 and various P_u (symbols) with the simulated results using the present model with (solid lines) or without (dash lines) updated rate constant of H + O_2 = O + OH by Varga et al. [2].

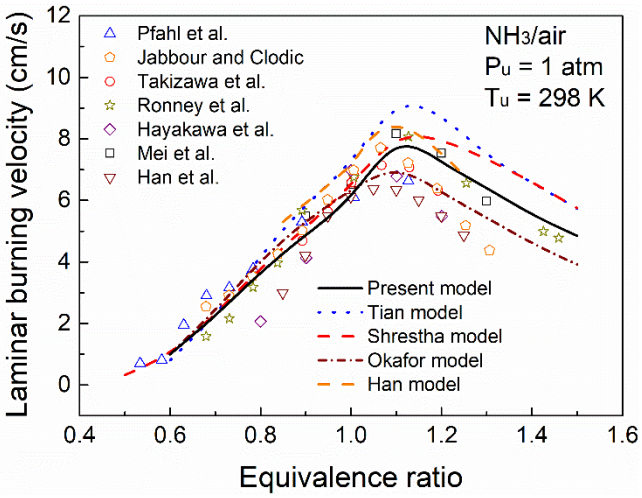


Figure S3 Validation against laminar burning velocity (LBV) data of NH_3/air mixtures at T_u = 298 K and P_u = 1 atm. Symbols denote the measured results in literature [3-9]. Lines denote the simulated results of the present model and previous models [10-13].

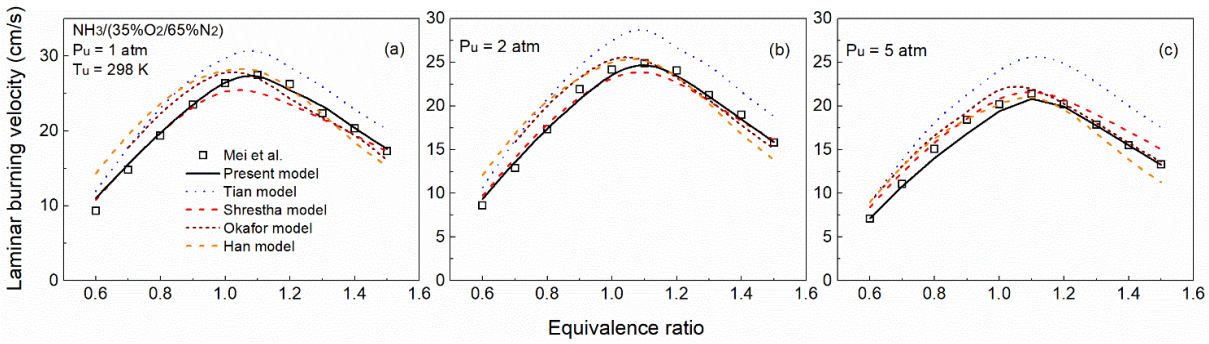


Figure S4 Validation against LBV data of NH_3/(35%O_2/65%N_2) mixtures at T_u = 298 K and P_u = 1-5 atm. Symbols denote the measured results in literature [9]. Lines denote the simulated results of the present model and previous models [10-13].

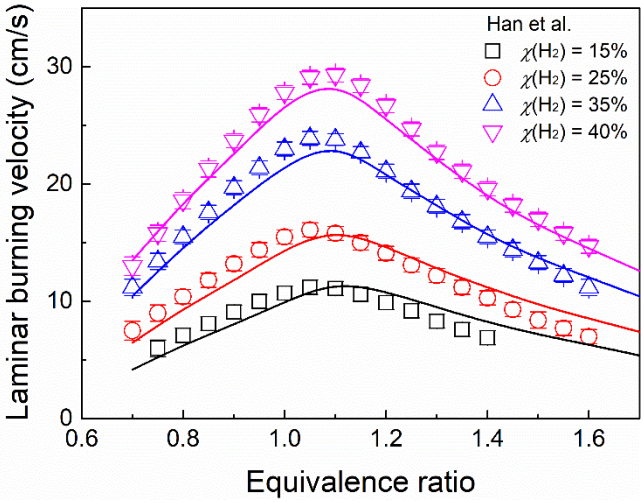


Figure S5 Validation against LBV data of NH_3/H_2/air mixtures with $\chi(H_2)$ from 15% to 40% at P_u = 1 atm. Symbols denote the experimental data measured by Han et al. [8]. Lines denote the simulated results of the present model.

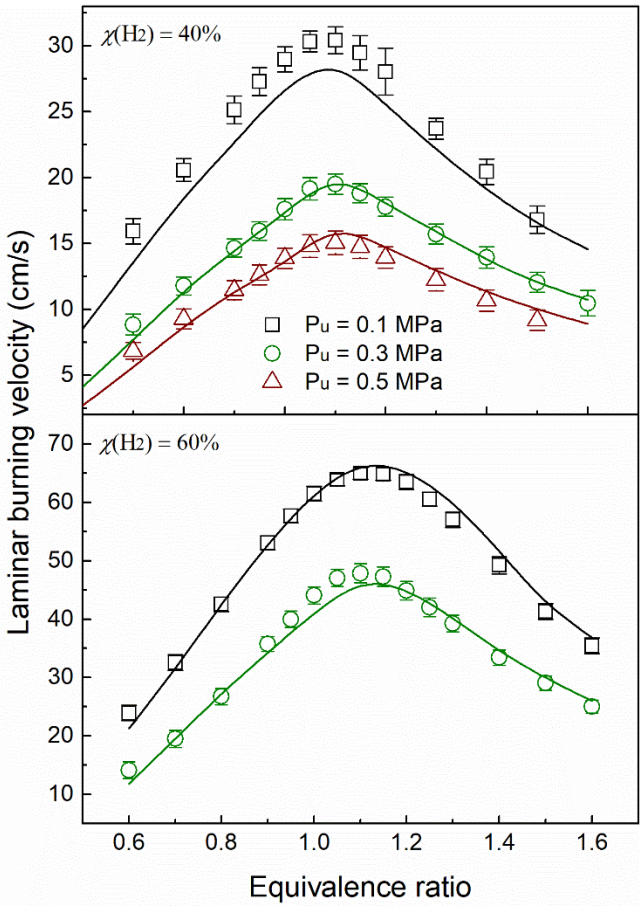


Figure S6 Validation against LBV data of NH_3/H_2/air mixtures with $\chi(H_2)$ = 40% and 60% at P_u = 1-5 atm. Symbols denote the experimental data measured by Wang et al. [14]. Lines denote the simulated results of the present model.

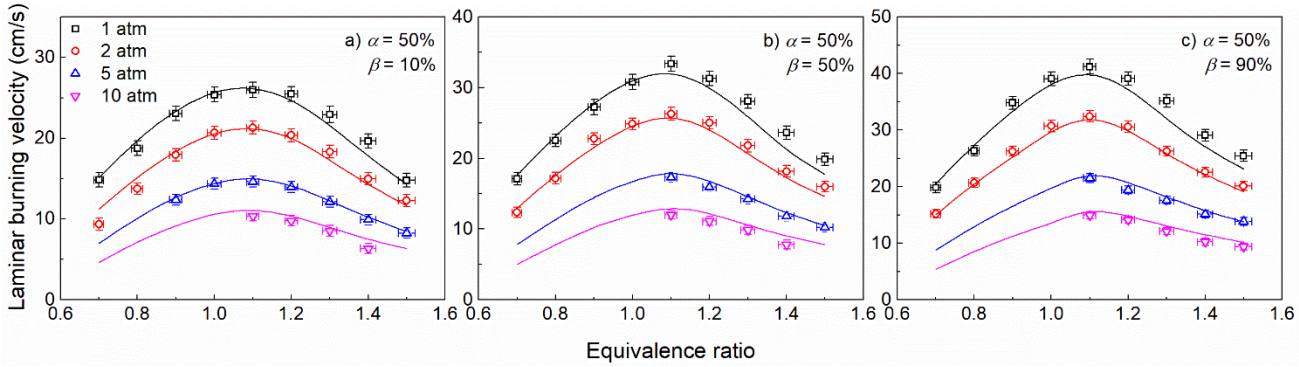


Figure S7 Validation against LBV data of NH_3/syngas/air mixtures with α = 50% and β = 10%, 50% and 90%. α and β denote the mole fraction of NH_3 in fuel mixture and that of H_2 in syngas mixture, respectively. Symbols denote the experimental data measured in our previous work [1]. Lines denote the simulated results of the present model.

S2. Validation against laminar premixed flame speciation data in literature

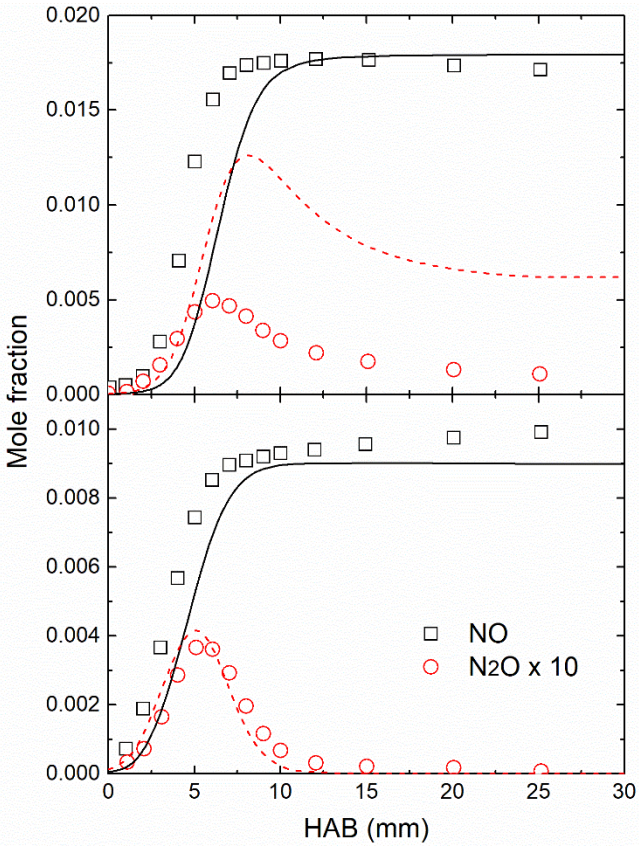


Figure S8 Validation against mole fraction profiles of NO and N_2O in (a) laminar premixed flame of 3.4%NH_3/13.5%H_2/76.5%O_2/6.6%Ar mixture at 34.5 Torr and inlet velocity of 59.8 cm/s and (b) laminar premixed flame of 3%NH_3/20.9%H_2/12.7%O_2/63.4%Ar mixture at 34.5 Torr and inlet velocity = 65.4 cm/s. Symbols denote the experimental data measured by Bian et al. [15]. Lines denote the simulated results of the present model.

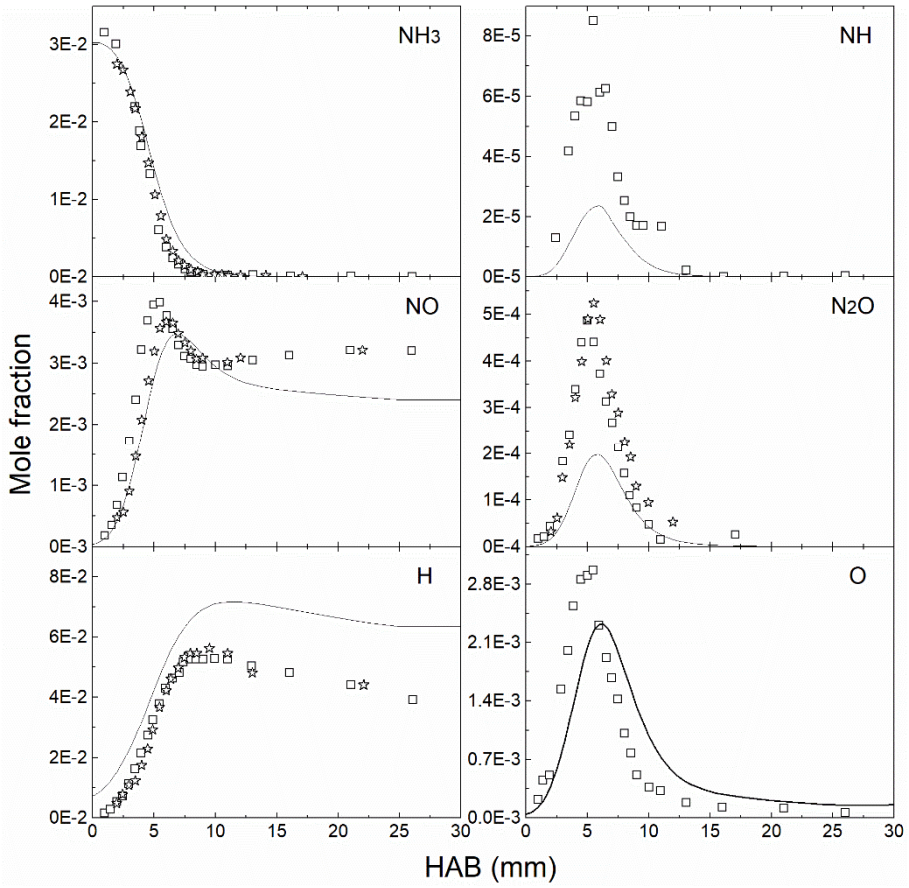


Figure S9 Validation against mole fraction profiles of species in laminar premixed 2.9%NH_3/35.4%H_2/10.4%O_2/51.3%Ar flame at 35.5 Torr and inlet velocity of 131 cm/s. Symbols denote the experimental data measured by Vandooren [16]. Lines denote the simulated results of the present model.

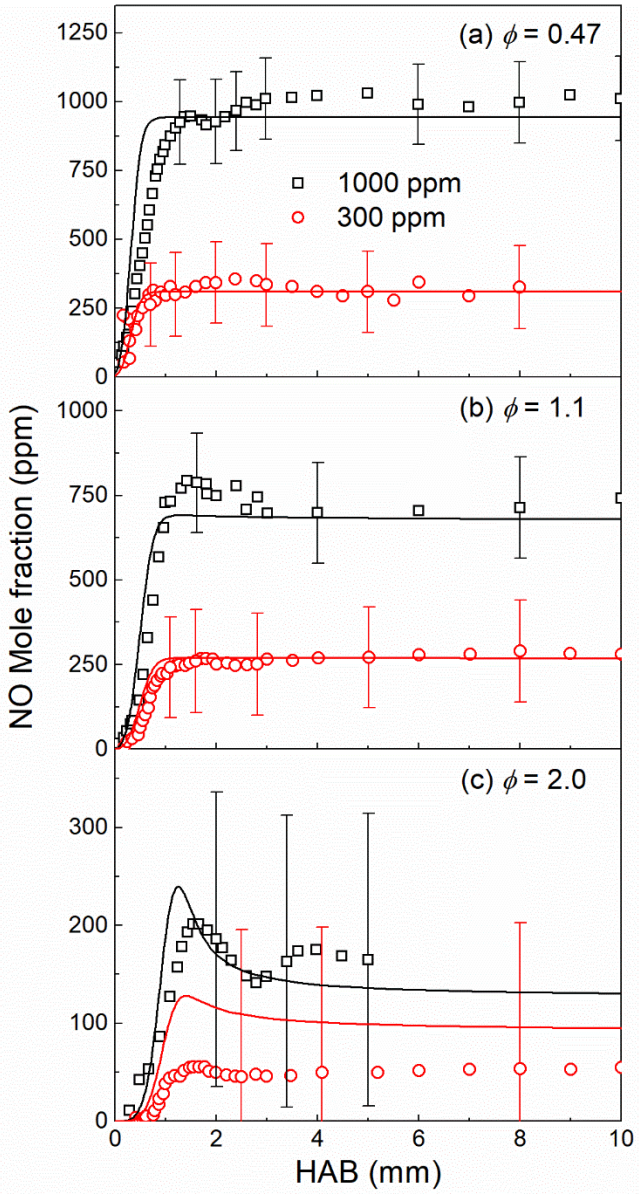


Figure S10 Validation against NO mole fraction profiles in laminar premixed $H_2/O_2/N_2$ flames doped with 300 (red) and 1000 (black) ppm NH_3 at atmospheric pressure and ϕ = (a) 0.47, (b) 1.1 and (c) 2.0. The volume fractions of O_2 in O_2 and N_2 mixtures are (a) 20.9%, (b) 9.0% and (c) 7.7%. Symbols denote the experimental data measured by Shmakov et al. [17]. Lines denote the simulated results of the present model.

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